Technology Stack & System Prerequisites

Project: NeuroClass — Multimodal Attendance & Engagement Agent

Module: `2.Requirement Analysis`

Phase: Technical Stack Architecture & Dependency Specifications

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1. Architectural Stack Overview

The NeuroClass system is built upon a high-performance, modular Python stack engineered specifically for **low-latency edge execution, hardware agnosticism, and 100% offline privacy**.

NEUROCLASS TECHNOLOGY STACK
PRESENTATION LAYER : Vanilla HTML5/CSS3/ES6 Modern HUD + Streamlit 1.65.0 Analytics Suite
GATEWAY & TELEMETRY : FastAPI 0.143.0 + Uvicorn 0.54.0 + WebSockets 17.2 (Async Event Loop)
COMPUTER VISION ENGINE : OpenCV 5.0.0 + MediaPipe 1.1.0 + DeepSort Realtime 1.3.2 Spatial Track
ACOUSTIC SIGNAL ENGINE : Librosa 1.0.0 + SoundFile 0.14.0 + SciPy 1.18.1 (VAD & Rhythm Dynamics)
FUSION & AGENT ENGINE : NumPy 2.5.4 + Pandas 3.0.6 + Groq 1.7.0 / LiteLLM Rule-Based Agent
PERSISTENCE & STORAGE : SQLite3 (ACID WAL Mode) + SQLAlchemy 2.1.4 Object Relational Mapping

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2. The 14 Core Dependencies Matrix

Every library in the core requirements has been verified through automated environment testing (`validate_environment.py`) with zero import conflicts or missing binary symbols:

#	Dependency	Verified Version	Core Architectural Purpose	License
1	`fastapi`	`0.143.0`	Asynchronous high-performance REST, SSE, and WebSocket API gateway	MIT
2	`uvicorn`	`0.54.0`	Lightning-fast ASGI web server implementation	BSD-3
3	`opencv-python`	`5.0.0`	Real-time frame capture, Haar face detection, ROI privacy blur, drawing	Apache-2.0
4	`mediapipe`	`1.1.0`	High-fidelity face mesh landmarking and 3D head pose estimation	Apache-2.0
5	`numpy`	`2.5.4`	High-throughput vectorized matrix math, coordinate geometry, RMS	BSD-3
6	`pandas`	`3.0.6`	Tabular data manipulation, time-series bucketing, CSV export formatting	BSD-3
7	`python-multipart`	`0.0.32`	Asynchronous file upload handling for 3-Way Hybrid Student Enrollment	Apache-2.0
8	`websockets`	`17.2`	Bi-directional full-duplex real-time telemetry streaming to client HUD	BSD-3
9	`sqlalchemy`	`2.1.4`	Enterprise-grade ORM and database schema migration abstraction	MIT
10	`librosa`	`1.0.0`	Acoustic feature extraction (spectral centroid, rolloff, zero-crossing)	ISC
11	`soundfile`	`0.14.0`	Uncompressed PCM WAV/FLAC audio ingestion and decoding	BSD-3
12	`groq`	`1.7.0`	Ultra-low latency LPU-accelerated pedagogical reasoning (optional cloud)	Apache-2.0
13	`deep-sort-realtime`	`1.3.2`	Multi-target Kalman-filter and cosine appearance metric tracking	MIT
14	`python-dotenv`	`1.2.4`	Local 12-factor environment configuration and secret management	BSD-3

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3. Preserved Runtime Enhancements Matrix

In addition to the 14 core requirements, the following runtime packages provide essential tooling, UI richness, and testing rigor:

Enhancement	Verified Version	Functional Justification
`streamlit`	`1.65.0`	Provides the interactive companion analytics dashboard and audit charts.
`requests`	`2.34.2`	Handles HTTP client requests and inter-service communication.
`pydantic`	`2.14.1`	Enforces runtime data validation and schema integrity on API payloads.
`scipy`	`1.18.1`	Powers signal processing filters and spatial distance matrices.
`matplotlib`	`3.11.2`	Renders downloadable static engagement heatmaps and focus curves.
`pytest`	`9.1.1`	Advanced test runner for continuous integration and automated test suites.
`reportlab`	`5.0.1`	Generates official, submission-grade PDF reports from project deliverables.

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4. Hardware & Operating System Prerequisites

NeuroClass is architected to operate efficiently across standard commodity personal computers, classroom workstations, and edge embedded devices without requiring discrete high-end GPUs or proprietary accelerator chips.

Minimum Hardware Specifications

- **Processor:** Quad-Core Intel Core i3 / AMD Ryzen 3 or equivalent (ARM64 supported).
- **RAM:** 4 GB available system memory.
- **Camera:** Standard 720p or 1080p USB webcam (or integrated laptop camera).
- **Microphone:** Integrated laptop mic or standard USB microphone.
- **Storage:** 500 MB free disk space for repository, virtual environment, and local DB.

Recommended Hardware Specifications

- **Processor:** 6-Core Intel Core i5 / AMD Ryzen 5 or Apple Silicon (M1/M2/M3).
- **RAM:** 8 GB or 16 GB system memory.
- **Acceleration:** Standard CPU execution (No discrete Nvidia GPU required).

Operating System Compatibility

- **Windows:** Windows 10 / Windows 11 (tested on Windows 11 Build 26300).
- **Linux:** Ubuntu 20.04 LTS / 22.04 LTS, Debian 11/12, Fedora 38+.
- **macOS:** macOS Monterey (12.0) through macOS Sonoma (14.0+).

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5. Security & Isolation Verification

To guarantee that no conflicting system packages contaminate runtime execution:

1. All dependencies are isolated inside a local virtual environment (`.venv`).
2. Verification script `validate_environment.py` executes before deployment to validate:

- Python version compatibility (***ge 3.10***).
- Virtual environment prefix isolation (`sys.prefix != sys.base_prefix`).
- Clean importation of all 14 libraries and enhancements.